STUDENT No

ANSWER SHEET

PROBLEM E1

GPhO Riyadh 2016

Problem E1. Rolling cylinder *(7 points)*

Part A. Critical slopes *(1 points)*

For critical slope α_0, directly measured quantities and calculations:

The critical slope $\alpha_0 =$

For critical slope α_2, directly measured quantities and calculations:

The critical slope $\alpha_2 =$

Part B. Rolling speed *(3 points)*

Rolling distance for average speed $l_t =$

rolling height h (mm)	rolling time, 1st segm.: t_1 (s)	speed, 1st segm.: v_1 (mm/s)	rolling time, 2nd segm.: t_2 (s)	speed, 2nd segm. (mm/s)	rolling time, long t_l (s)	average speed: v_l (mm/s)

The critical slope $\alpha_1 =$

Part C. Force as a function of speed *(2.3 points)*

Formula used to calculate the force:

Rolling height h (mm)	Rolling speed v (mm/s)	Force applied (F_m): (mN)				

Graph: F_m versus v

Suggested formula for the dependence $F_m(v)$:
(Indicate the values of all the parameters of this dependence.)

Part D. Mass of liquid *(0.7 points)*

Formula used to calculate the mass of liquid:

Mass of liquid $m =$